

Simão Pedro Sá Cunha

PhD Candidate in Informatics

 simaocunha71  simaocunha71  simao.s.cunha@inesctec.pt  simaocunha71.github.io

EDUCATION

- Doctoral Program in Informatics (PhD)** *Braga, Portugal*
University of Minho *2025 - Present*
- Informatics Engineering (Integrated Master's)** *Braga, Portugal*
University of Minho *2019 - 2024*
- Basic and Secondary Education** *Vila Nova de Famalicão, Portugal*
Escola Cooperativa Vale S. Cosme - Didáxis *2012 - 2019*

WORK EXPERIENCE

- Research Assistant** *Sep 2025 - Present*
HASLab, INESC TEC *Braga, Portugal*
 - Research activities during the PhD program.
- Teaching Assistant** *Feb 2025 - Jul 2026*
University of Minho *Braga, Portugal*
 - Courses taught:
 - * Imperative Programming (2025/2026)
 - * Topics of Software Development (2024/2025, 2025/2026)
 - * Experimentation in Software Engineering (2024/2025)
- Summer Internship** *Jul - Sep 2023*
Bosch Portugal *Braga, Portugal*
 - Applied machine learning and deep learning techniques to develop a system for detecting delaminations.
 - Implemented machine learning models using Python libraries such as Scikit-learn (Sklearn) and deep learning models using Deep Learning Halcon by MVTEC to build robust detection algorithms.
 - Conducted image processing tasks using Halcon from MVTEC to enhance data preprocessing.
 - Contributed to internal software development projects.
 - Gained foundational knowledge in computer vision.
- Summer Internship** *Jul - Aug 2022*
Accenture Portugal *Braga, Portugal*
 - Introduction to Google Cloud Platform.
 - Data Fusion instance creation and JSON file upload with a customized pipeline configuration in a Cloud Build file written in YAML.
 - Agile Scrum Methodology applied in the entire internship.

LANGUAGES

Portuguese: Native **English:** C1 - Advanced

SKILLS

Programming Languages: C, Java, JavaScript, Python, Haskell
Frontend Frameworks: React Native
Energy Monitoring: Intel RAPL
Cloud and DevOps: Google Cloud Platform, Firebase, Docker, Ansible
Business Intelligence: Microsoft PowerBI
Research Interests: Large Language Models, Machine Learning, Green Software

PUBLICATIONS

- Luís Maia, Simão Cunha and João Saraiva, **Why Just-In-Time Compilation Matters: Evaluating Runtime and Energy Efficiency**, in *SLE '26: Proceedings of the 19th ACM SIGPLAN International Conference on Software Language Engineering*, 2026. <https://dl.acm.org/doi/10.1145/3806383.3815520>
- Simão Cunha, Francisco Ribeiro, Luís Cruz and João Saraiva, **Not All Local LLMs Are Equal: A Benchmark of Energy and Performance**, in GREENS '26 at ICSE 2026 (to be published)
- André Brandão, Diogo Matos, Miguel Guimarães, Simão Cunha and João Saraiva, **The Green Side of the Lua**, in *2026 IEEE International Conference on Software Analysis, Evolution and Reengineering - Companion (SANER-C)*. <https://ieeexplore.ieee.org/document/11500263>
- Luís Cruz, João Paulo Fernandes, Maja Hanne Kirkeby, Silverio Martínez-Fernández, June Sallou, Hina Anwar, Enrique Barba Roque, Justus Bogner, Joel Castaño, Fernando Castor, Aadil Chasmawala, Simão Cunha, Daniel Feitosa, Alexandra González, Andreas Jedlitschka, Patricia Lago, Ana Oprescu, Pooja Rani, João Saraiva, Federica Sarro, Raghavendra Selvan, Karthik Vaidhyanathan, Roberto Verdecchia, Ivan P. Yamshchikov, & Henry Muccini. 2025. **Greening AI-enabled systems with software engineering: A research agenda for environmentally sustainable AI practices**, in *SIGSOFT Software Engineering Notes (ACM)*, 2025. <https://doi.org/10.1145/3743095.3743099>
- Simão Cunha, Luís Silva, João Saraiva, and João Paulo Fernandes. 2024. **Trading Runtime for Energy Efficiency: Leveraging Power Caps to Save Energy Across Programming Languages**. *Proceedings of the 17th ACM SIGPLAN International Conference on Software Language Engineering (SLE '24)*, October 20–21, 2024, Pasadena, CA, USA. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3687997.3695638>
- Luís Maia, Marta Sá, Inês Ferreira, Simão Cunha, Luís Silva, Paulo Azevedo, and João Saraiva. 2024. **On the Impact of PowerCap in Haskell, Java, and Python**. *RAW'24: 3rd Workshop on Resource Awareness of Systems and Society*, July 02–05, 2024, Maribor, Slovenia. <https://ceur-ws.org/Vol-3867/paper4.pdf>

CERTIFICATES

- **10th International Workshop on Green and Sustainable Software (GREENS '26)** Apr 2026
ACM & IEEE & SIGSOFT Rio de Janeiro, Brazil
– Attendance Certificate
- **1st International Workshop on Green Software Evolution (Greenvolve 2026)** Mar 2026
University of Cyprus & IEEE & EasyConferences Online
– Attendance Certificate
- **Symposium on Sustainable IT Systems (SUITS)** Set 2025
DIREC & NNF & Roskilde U. & CPH U. & SDU & IT-Univ. & TUD & Aalborg U. Copenhagen, Denmark
– Attendance Certificate
- **CECAM-Lorentz workshop "Greening AI with Software Engineering"** Feb 2025
CECAM-HQ EPFL Lausanne, Switzerland
– Attendance Certificate
- **Cerciras Training School 2024** Aug 2025
University of Klagenfurt Klagenfurt, Austria
– Attendance Certificate
- **1st UMinho Research & Innovation Open Days 2024** Jan 2024
University of Minho Braga, Portugal
– Pitch Session Certificate
– Poster Session Certificate
- **Sustainable Summer School 2023** Jul 2023
University of Coimbra Coimbra, Portugal
– Attendance Certificate
– Students Workshop and Poster Session Certificate
- **English C1** Oct 2021
BabeliUM Braga, Portugal
– Certificate